Rainfall Estimation

This note summarizes results for a collection of 100 patches. All aggregate figures and tables that follow should therefore be read as patch-level summaries over this hundred-patch evaluation set, rather than as statistics from a single patch or from the full archive of available patches.

Relative Rainy-Pixel Distance Profile

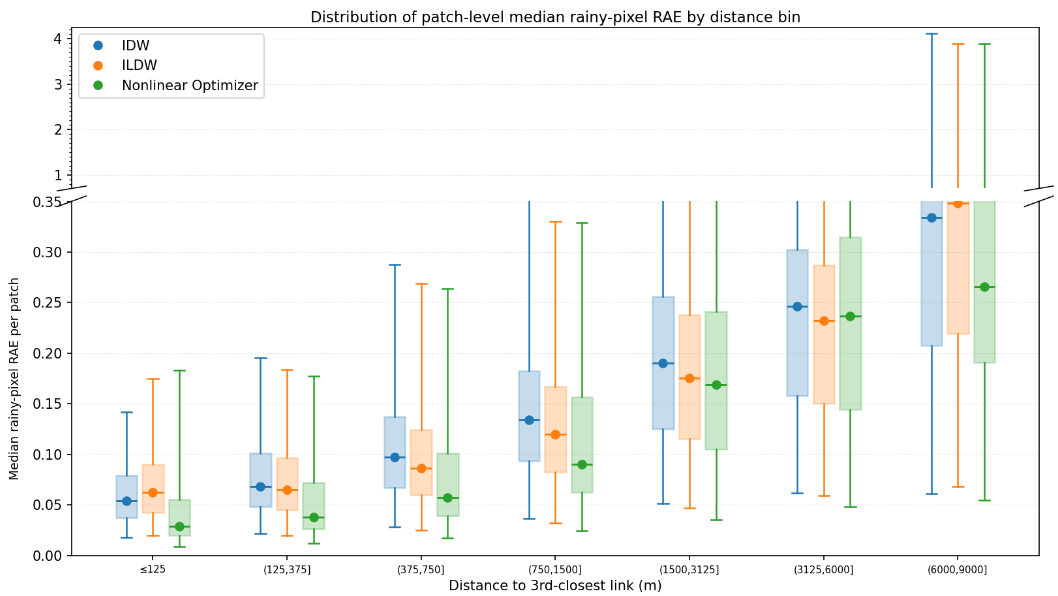

Figure 1: Rainy-pixel distance-profile box-and-whisker summary for distance to the third closest link. Each patch contributes one median rainy-pixel relative-absolute-error (RAE) value per distance bin, and these patch-level medians are then summarized across patches with a box-and-whisker plot. Here, a rainy pixel is a pixel whose ground-truth rainfall is at least the configured wet threshold (set to 0.6), and relative absolute error for a rainy pixel p is defined as $|GT(p) - PRED(p)|/GT(p)$.

This plot summarizes how solver performance changes as pixels get farther from the third-nearest link. Lower values indicate lower rainy-pixel relative absolute error in that distance bin.

Rainy-pixel Count Table for Distance to the 3rd Closest Link

The table below gives the average number of rainy pixels per patch and the corresponding standard deviation for each distance bin to the 3rd closest link. These are the same patch-level counts summarized in the tick labels of the rainy-pixel distance-profile plots.

Distance bin (to the 3rd closest link)	Avg. rainy pixels / patch	SD
≤ 125 m	4267.0	960.4
$(125, 375]$ m	16737.7	3803.2
$(375, 750]$ m	26809.9	6707.0
$(750, 1500]$ m	51988.5	14468.0
$(1500, 3125]$ m	88839.4	27531.1
$(3125, 6000]$ m	45929.4	15679.4
$(6000, 9000]$ m	3835.8	2725.7
> 9000 m	514.4	966.9

Table 1: Rainy-pixel counts per patch for the distance bins to the 3rd closest link used in the rainy-pixel distance-profile plots.

Confusion Map

IDW

GT	Prediction	
	Wet	Dry
Wet	0.893 ± 0.013	0.002 ± 0.000
Dry	0.040 ± 0.004	0.065 ± 0.010

ILDW

GT	Prediction	
	Wet	Dry
Wet	0.892 ± 0.013	0.003 ± 0.000
Dry	0.037 ± 0.004	0.068 ± 0.010

Nonlinear Optimizer

GT	Prediction	
	Wet	Dry
Wet	0.890 ± 0.013	0.005 ± 0.001
Dry	0.030 ± 0.004	0.075 ± 0.010

Table 2: Patch-averaged wet/dry confusion matrices for IDW, ILDW, and the nonlinear optimizer at a threshold of 0.6 mm/h, where a pixel is called wet if its rainfall is at least 0.6 mm/h and dry otherwise. For each patch, pixels are first assigned to the four confusion categories defined by the ground-truth and predicted wet/dry labels: top-left is TP, top-right is FN, bottom-left is FP, and bottom-right is TN. The count in each category is then divided by the total number of pixels in that patch, producing a patch-level pixel fraction. The table entries report the mean of those patch-level fractions across the 100 evaluated patches, together with the standard error of the mean (SEM). These are therefore fractions of all pixels in a patch, not row-normalized conditional rates.

Patch Geometry Overview

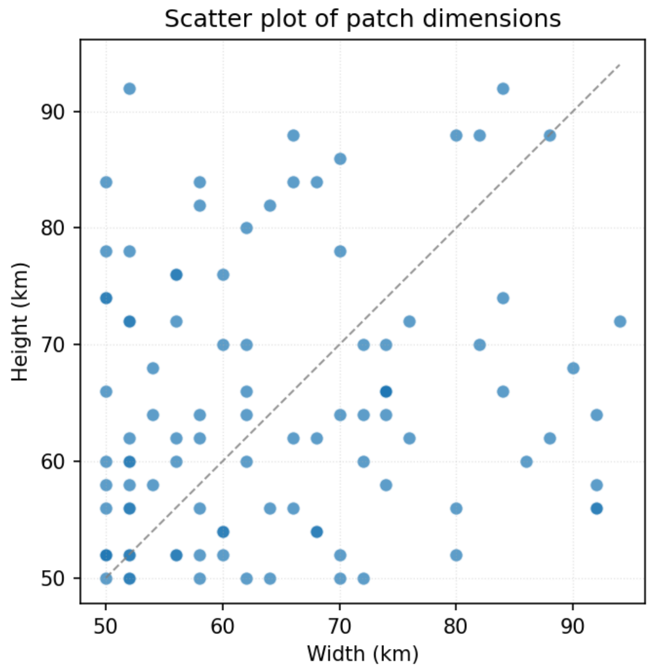

Figure 2: Scatter plot of patch dimensions for the 100 patches used in the analysis, with one marker per patch. The figure summarizes the range of patch geometries represented in the dataset and makes it clear that the evaluation is not restricted to a single fixed patch shape. The dashed diagonal is the 1:1 line between width and height.

This figure makes it easy to see whether the selected patches occupy a tight geometry range or whether the report is mixing very different patch extents.

Link Count Distribution

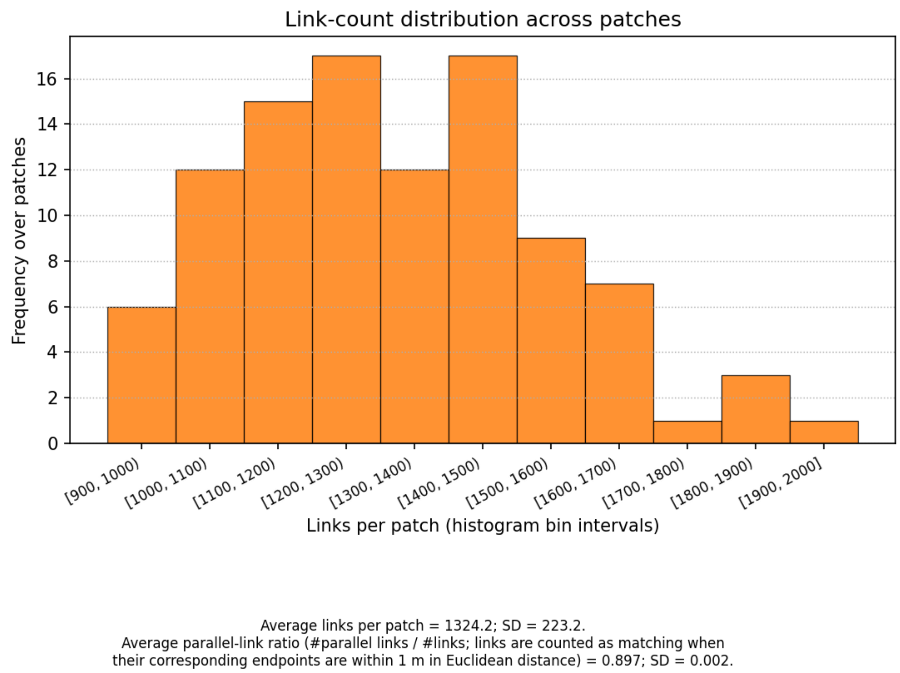

Figure 3: Histogram of the number of links per patch. The figure footnote also reports the average parallel-link ratio, defined here as the fraction of links in a patch that have at least one parallel partner.

This histogram summarizes how link density varies across the selected patches. The parallel-link ratio is included because the current expectation is that it should stay close to 1 for this dataset.

Approx. patch size (km)	# links	1-multiplicity	2-multiplicity	3-multiplicity
50×50	939	99	840	0
62×60	1254	128	1126	0
72×70	1507	157	1350	0

Table 3: Representative link multiplicities for three HundredPatches cases whose dimensions are close to 50×50, 60×60, and 70×70 km. Here, multiplicity is defined by grouping links whose corresponding endpoints match within 1 m in Euclidean distance, allowing endpoint order reversal. A 1-multiplicity link appears alone, a 2-multiplicity link belongs to an exactly duplicated pair, and a 3-multiplicity link belongs to a triplet of coincident links. The entries in the table count links, not multiplicity groups.